

ORF 524 – Midterm Exam

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Notes:

- Duration of examination: *4 hours*.
Additional 20 minutes are given to upload your solutions to Canvas.
- All parts are equally weighted (5 points).
Please provide as much detail as possible in your answers.
Answers without proper justification will not receive (partial) credit.
- Please start each question on a separate page.
A single PDF file must be submitted to Canvas.
- Only two pages of personal notes are allowed during the examination.
Basic auxiliary information about standard distributions is provided at the end of the exam.
This is otherwise a closed-book and individual examination.
Do not consult textbooks, online materials, or other sources.
Do not consult with classmates, other people or in online forums.

1 Question 1: Probability Theory (25 points)

Let $(X_1, X_2, \dots, X_n)$ be i.i.d. copies of an integrable random variable $X \sim \mathbb{P}$, with $M(t) = \mathbb{E}[\exp(tX)] < \infty$ for $t \geq 0$. Recall that $X_{(n)} = \max_{1 \leq i \leq n} X_i$, and that parametrizations of basic distributions are given at the end of the exam.

1. Show that

$$\exp(t\mathbb{E}[X_{(n)}]) \leq nM(t), \quad t \geq 0.$$

• **Solution:** for all $t > 0$, we have:

$$\begin{aligned} \exp(t\mathbb{E}[X_{(n)}]) &\leq \mathbb{E} \left[\exp \left(t \max_{1 \leq i \leq n} X_i \right) \right] = \mathbb{E} \left[\max_{1 \leq i \leq n} \exp(tX_i) \right] \\ &\leq \mathbb{E} \left[\sum_{i=1}^n \exp(tX_i) \right] = \sum_{i=1}^n \mathbb{E}[\exp(tX_i)] = nM(t) \end{aligned}$$

where the first inequality uses Jensen inequality.

2. Show that

$$\mathbb{E}[X_{(n)}] \leq \inf_{t>0} \frac{\log n + \log M(t)}{t}.$$

• **Solution:** Using part 1, after taking log both sides, the generic maximal bound follows directly because the bound in part 1 holds for all $t > 0$.

3. Let $\mathbb{P} = \mathcal{N}(0, \sigma^2)$. Show that

$$\mathbb{E}[X_{(n)}] \leq \sqrt{2}\sigma\sqrt{\log n}.$$

• **Solution:** Recall that the MGF of a Gaussian random variable is $M(t) = \exp(\mu t + \sigma^2 t^2/2)$, which gives

$$\mathbb{E}[X_{(n)}] \leq \inf_{t>0} \left\{ \frac{\log n}{t} + \frac{t\sigma^2}{2} \right\}$$

and the result follows.

4. Let $\mathbb{P} = \text{Gamma}(\alpha, \beta)$. Show that, for $n \geq 2$,

$$\mathbb{E}[X_{(n)}] \leq \frac{2^{1/\alpha}}{2^{1/\alpha} - 1} 2\beta \log n.$$

• **Solution:** For a Gamma random variable, $M(t) = (1 - \beta t)^{-\alpha}$. Then, using part 2 with $t = (1 - n^{-1/\alpha})/\beta$, we have

$$\mathbb{E}[X_{(n)}] \leq \frac{2\beta \log n}{1 - n^{-1/\alpha}},$$

and the result follows because the denominator is increasing in n .

5. Let $(Z_i : 1 \leq i \leq n)$ be i.i.d. $N(0, \sigma^2)$ and $(E_i : 1 \leq i \leq n)$ be i.i.d. $\text{Exponential}(0, \theta)$, not necessarily independent. Show that, for $n \geq 2$,

$$\mathbb{E} \left[\max_{1 \leq i \leq n} |Z_i| \sqrt{E_i} \right] \leq \frac{8}{\sqrt{3}} \sigma \sqrt{\theta} \log n.$$

- **Solution:** Using Cauchy-Schwarz inequality and part 4,

$$\mathbb{E} \left[\max_{1 \leq i \leq n} |Z_i| \sqrt{E_i} \right] \leq \sqrt{\mathbb{E} \left[\max_{1 \leq i \leq n} Z_i^2 \right]} \sqrt{\mathbb{E} \left[\max_{1 \leq i \leq n} E_i \right]} \leq \sqrt{\frac{4\sigma^2 \log n}{(1 - n^{-2})}} \sqrt{\frac{2\theta \log n}{(1 - n^{-1})}}$$

because $(Z_i/\sigma)^2 \sim \chi_1^2 = \text{Gamma}(1/2, 2)$ and $E_i \sim \text{Exponential}(0, \theta) = \text{Gamma}(1, \theta)$.

2 Question 2: Average Risk Optimality (25 points)

Let $\mathbf{X} \sim \mathbb{P}_\theta$ be a random vector with $\mathbb{P}_\theta \in \mathcal{P} = \{\mathbb{P}_\theta : \theta \in \Theta\}$. Recall that the risk of a statistic $T(\mathbf{X})$ for the parameter $\Psi(\theta)$ under square loss is $R(\Psi(\theta), T) = \mathbb{E}_\theta[(T(\mathbf{X}) - \Psi(\theta))^2]$.

1. Show that, in general, there is no unique solution (over $\mathcal{P}$) to the optimization problem:

$$\min_{T \in \mathcal{T}} \mathbb{E}_\theta[(T(\mathbf{X}) - \Psi(\theta))^2], \quad \mathcal{T} = \{T(\cdot) : \mathbb{E}_\theta[|T(\mathbf{X})|^2] < \infty\}.$$

- **Solution:** The solution is not unique because the objective function can be made equal to zero by setting $T(\mathbf{X}) = \Psi(\theta)$. Thus, the solution depends on the value of θ .

2. One way to resolve the lack of uniqueness is to restrict the class of statistics $T(\cdot)$ considered (e.g., allow for unbiased estimators only, that is, restrict to $\mathcal{T}_U$). An alternative solution is to consider *Average Risk*: assume the parameter θ is a random variable with Lebesgue density $f(\theta)$, sometimes called a *prior*, and consider the new optimization problem:

$$\min_{T \in \mathcal{T}_B} \int_{\Theta} \mathbb{E}_\theta[(T(\mathbf{X}) - \Psi(\theta))^2] f(\theta) d\theta, \quad \mathcal{T}_B = \left\{ T(\cdot) : \int_{\Theta} \mathbb{E}_\theta[|T(\mathbf{X})|^2] f(\theta) d\theta < \infty \right\},$$

with $\mathbb{E}_\theta[\cdot]$ now denoting the expectation with respect to the conditional distribution of $\mathbf{X}|\theta$. Show that there exists a *unique* (a.s.) solution to this problem, denoted by $T^*(\mathbf{X})$, and give its precise expression. This optimal procedure is called the *Bayes rule* (for prior $f(\theta)$).

- **Solution:** By the LIE, the objective function can be rewritten as:

$$\mathbb{E}[(T(\mathbf{X}) - \Psi(\theta))^2]$$

where the expectation is taken with respect to the joint density of $\mathbf{X}$ and θ . Relabel $\Psi(\theta) := Y$ so that the problem is simply:

$$\min_{T \in \mathcal{T}_B} \mathbb{E}[(Y - T(\mathbf{X}))^2],$$

which was solved in class. The solution is $T^*(\mathbf{X}) = \mathbb{E}[Y|\mathbf{X}] = \mathbb{E}[\Psi(\theta)|\mathbf{X}]$.

3. Assume $\mathbf{X}$ is a continuous random variable (only for simplicity). Show that

$$T^*(\mathbf{X}) = \frac{\int_{\Theta} \Psi(\theta) f(\mathbf{X}|\theta) f(\theta) d\theta}{\int_{\Theta} f(\mathbf{X}|\theta) f(\theta) d\theta},$$

where $f(\mathbf{X}|\theta)$ denotes the Lebesgue density of the conditional distribution of $\mathbf{X}|\theta$, and $f(\theta|\mathbf{X})$ is interpreted as the Lebesgue density of the (posterior) distribution of $\theta|\mathbf{X}$.

• **Solution:** We have that

$$\begin{aligned} T^*(\mathbf{X}) &= \int_{\Theta} \Psi(\theta) f(\theta|\mathbf{X}) d\theta = \int_{\Theta} \Psi(\theta) \frac{f(\mathbf{X}|\theta) f(\theta)}{f(\mathbf{X})} d\theta = \frac{\int_{\Theta} \Psi(\theta) f(\mathbf{X}|\theta) f(\theta) d\theta}{f(\mathbf{X})} \\ &= \frac{\int_{\Theta} \Psi(\theta) f(\mathbf{X}|\theta) f(\theta) d\theta}{\int_{\Theta} f(\mathbf{X}|\theta) f(\theta) d\theta} \end{aligned}$$

by Bayes' theorem.

4. Let $\mathbf{X} = (X_1, X_2, \dots, X_n)'$ i.i.d. $N(\theta, \sigma^2)$ conditionally on θ , with $\sigma^2 > 0$ known. In addition, impose a Gaussian prior for θ , that is, let $\theta \sim N(\mu, \tau^2)$, with $\mu \in \mathbb{R}$ and $\tau^2 > 0$.

(a) Show that $\theta|\mathbf{X} \sim N\left(\mu \cdot \frac{\sigma^2}{n\tau^2 + \sigma^2} + \bar{X}_n \cdot \frac{n\tau^2}{n\tau^2 + \sigma^2}, \frac{\sigma^2\tau^2}{n\tau^2 + \sigma^2}\right)$, where $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$.

Hint: for an invertible matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$ and any vector $\mathbf{u} \in \mathbb{R}^n$ such that $\mathbf{u}'\mathbf{A}^{-1}\mathbf{u} \neq -1$,

$$(\mathbf{A} + \mathbf{u}\mathbf{u}')^{-1} = \mathbf{A}^{-1} - \frac{\mathbf{A}^{-1}\mathbf{u}\mathbf{u}'\mathbf{A}^{-1}}{1 + \mathbf{u}'\mathbf{A}^{-1}\mathbf{u}}.$$

• **Solution:** First, recall that if $\mathbf{Y} \in \mathbb{R}^d$ and $\mathbf{W} \in \mathbb{R}^k$ with

$$\begin{bmatrix} \mathbf{Y} \\ \mathbf{W} \end{bmatrix} \sim N_{d+k}(\boldsymbol{\mu}, \boldsymbol{\Sigma}), \quad \boldsymbol{\mu} = \begin{bmatrix} \boldsymbol{\mu}_Y \\ \boldsymbol{\mu}_W \end{bmatrix} \in \mathbb{R}^{d+k}, \quad \boldsymbol{\Sigma} = \begin{bmatrix} \boldsymbol{\Sigma}_{YY} & \boldsymbol{\Sigma}_{YW} \\ \boldsymbol{\Sigma}_{YW} & \boldsymbol{\Sigma}_{WW} \end{bmatrix} \in \mathbb{R}^{d \times k},$$

then

$$\mathbf{Y}|\mathbf{W} \sim N_d\left(\boldsymbol{\mu}_Y + (\mathbf{W} - \boldsymbol{\mu}_W)' \boldsymbol{\Sigma}_{WW}^{-1} \boldsymbol{\Sigma}_{WY}, \boldsymbol{\Sigma}_{YY} - \boldsymbol{\Sigma}_{YW} \boldsymbol{\Sigma}_{WW}^{-1} \boldsymbol{\Sigma}_{WY}\right).$$

Next, note that because $\mathbf{X}|\theta$ is Gaussian with constant variance, and θ is also Gaussian, then so is $(\mathbf{X}', \theta)'$. (This is not true in general.) Thus, we conclude that

$$\begin{bmatrix} \theta \\ \mathbf{X} \end{bmatrix} \sim N_{n+1}\left(\begin{bmatrix} \mu \\ \mu \mathbf{1}_n \end{bmatrix}, \begin{bmatrix} \tau^2 & \tau^2 \mathbf{1}_n' \\ \tau^2 \mathbf{1}_n' & \tau^2 \mathbf{1}_n \mathbf{1}_n' + \sigma^2 \mathbf{I}_n \end{bmatrix}\right),$$

where $\mathbf{1}_n = (1, 1, \dots, 1)'$ denotes the n -dimensional column vector of ones and $\mathbf{I}_n$ denotes the n -dimensional identity matrix, and because

$$\begin{aligned} \mu_{\mathbf{X}} &= \mathbb{E}[\mathbf{X}] = \mathbb{E}[\mathbb{E}[\mathbf{X}|\theta]] = \mathbb{E}[\theta] \mathbf{1}_n = \mu \mathbf{1}_n, \\ \Sigma_{\mathbf{X}\theta} &= \text{Cov}[\mathbf{X}, \theta] = \mathbb{E}[\mathbf{X}\theta] - \mathbb{E}[\mathbf{X}]\mathbb{E}[\theta] = \mathbb{E}[\theta^2] \mathbf{1}_n - \mu^2 \mathbf{1}_n = \tau^2 \mathbf{1}_n, \\ \Sigma_{\mathbf{X}\mathbf{X}} &= \mathbb{V}[\mathbf{X}] = \mathbb{V}[\mathbb{E}[\mathbf{X}|\theta]] + \mathbb{E}[\mathbb{V}[\mathbf{X}|\theta]] = \tau^2 \mathbf{1}_n \mathbf{1}_n' + \sigma^2 \mathbf{I}_n. \end{aligned}$$

Therefore, setting $(\mathbf{Y}', \mathbf{W}')' = (\theta, \mathbf{X}')'$, and noting that by the Sherman-Morrison formula

$$\Sigma_{\mathbf{X}\mathbf{X}}^{-1} = (\tau^2 \mathbf{1}_n \mathbf{1}_n' + \sigma^2 \mathbf{I}_n)^{-1} = \frac{1}{\sigma^2} \mathbf{I}_n - \frac{\tau^2}{n\tau^2\sigma^2 + \sigma^4} \mathbf{1}_n \mathbf{1}_n',$$

using standard matrix inverse results, we conclude that $\theta|\mathbf{X} \sim \mathcal{N}(\boldsymbol{\mu}_{\theta|X}, \boldsymbol{\Sigma}_{\theta|X})$ with

$$\begin{aligned}\boldsymbol{\mu}_{\theta|X} &= \boldsymbol{\mu}_\theta + (\mathbf{X} - \boldsymbol{\mu}_\mathbf{X})' \boldsymbol{\Sigma}_{\mathbf{X}\mathbf{X}}^{-1} \boldsymbol{\Sigma}_{\mathbf{X}\theta} \\ &= \mu + (\mathbf{X} - \mu \mathbf{1}_n)' \left(\frac{1}{\sigma^2} \mathbf{I}_n - \frac{\tau^2}{n\tau^2\sigma^2 + \sigma^4} \mathbf{1}_n \mathbf{1}_n' \right) \tau^2 \mathbf{1}_n \\ &= \mu + (n\bar{X}_n - n\mu) \frac{\tau^2}{n\tau^2 + \sigma^2} \\ &= \mu \cdot \frac{\sigma^2}{n\tau^2 + \sigma^2} + \bar{X}_n \cdot \frac{n\tau^2}{n\tau^2 + \sigma^2}\end{aligned}$$

because $\mathbf{1}_n' \mathbf{1}_n = n$ and $\mathbf{X}' \mathbf{1}_n = n\bar{X}_n$, and

$$\boldsymbol{\Sigma}_{\theta|X} = \boldsymbol{\Sigma}_{\theta\theta} - \boldsymbol{\Sigma}_{\theta\mathbf{X}} \boldsymbol{\Sigma}_{\mathbf{X}\mathbf{X}}^{-1} \boldsymbol{\Sigma}_{\mathbf{X}\theta} = \tau^2 - \frac{n\tau^4}{\sigma^2} + \frac{n^2\tau^6}{n\tau^2\sigma^2 + \sigma^4} = \frac{\sigma^2\tau^2}{n\tau^2 + \sigma^2}.$$

- (b) Give the form of the optimal Average Risk (or Bayes) estimator, and compute the associated (Bayes) Risk, when $\Psi(\theta) = \theta$ and when $\Psi(\theta) = \theta^2$.

Discuss what happens when the prior distribution, $f(\theta)$, becomes more concentrated (i.e., $\tau \rightarrow 0$). Explain whether the Bayes procedure for $\Psi(\theta) = \theta$, $\Psi(\theta) = \theta^2$ or other parameters becomes less or more dogmatic (i.e., more dominated by the prior chosen).

- **Solution:** The optimal Bayes estimator is $T^*(\mathbf{X}) = \mathbb{E}[\theta|\mathbf{X}] = \int_{\Theta} \Psi(\theta) f(\theta|\mathbf{X}) d\theta$, and its associated Bayes risk is

$$\mathbb{E}[(\Psi(\theta) - \mathbb{E}[\Psi(\theta)|\mathbf{X}])^2] = \mathbb{E}[\mathbb{E}[(\Psi(\theta) - \mathbb{E}[\Psi(\theta)|\mathbf{X}])^2|\mathbf{X}]] = \mathbb{E}[\mathbb{V}[\Psi(\theta)|\mathbf{X}]]$$

where the outer expectation is taken with respect to $f(\mathbf{X})$.

When $\Psi(\theta) = \theta$, the Bayes rule is

$$T^*(\mathbf{X}) = \mathbb{E}[\theta|\mathbf{X}] = \mu \cdot \frac{\sigma^2}{n\tau^2 + \sigma^2} + \bar{X}_n \cdot \frac{n\tau^2}{n\tau^2 + \sigma^2}$$

and its Bayes risk is

$$\mathbb{E}[\mathbb{V}[\theta|\mathbf{X}]] = \frac{\sigma^2\tau^2}{n\tau^2 + \sigma^2}.$$

Similarly, when $\Psi(\theta) = \theta^2$, the Bayes rule is

$$\begin{aligned}T^*(\mathbf{X}) &= \mathbb{E}[\theta^2|\mathbf{X}] = \mathbb{V}[\theta|\mathbf{X}] + (\mathbb{E}[\theta|\mathbf{X}])^2 \\ &= \frac{\sigma^2\tau^2}{n\tau^2 + \sigma^2} + \left(\mu \cdot \frac{\sigma^2}{n\tau^2 + \sigma^2} + \bar{X}_n \cdot \frac{n\tau^2}{n\tau^2 + \sigma^2} \right)^2,\end{aligned}$$

and its associate Bayes risk, $\mathbb{E}[\mathbb{V}[\theta^2|\mathbf{X}]]$, can be computed analogously.

Finally, consider what happens when the dispersion of the prior distribution decreases: $\tau \rightarrow 0$. For $\Psi(\theta) = \theta$, the mean of the posterior distribution converges to

μ , while the variance goes to 0. Hence, the Bayes estimator becomes the prior mean μ and the risk becomes 0. In other words, the Bayes procedure becomes more and more dogmatic as it is more and more driven by the prior distribution. A similar phenomena happens for the case of $\Psi(\theta) = \theta^2$. In general, the more concentrated the prior, the more dogmatic the (optimal) Bayes procedure.

3 Question 3: Finite-Sample Statistical Inference (50 points)

Let $(X_i : 1 \leq i \leq n)$ be i.i.d. with $X_i \sim \mathbb{P}_\theta$, $\mathbb{E}_\theta[X_i] = \sqrt{\theta}$, and Lebesgue density

$$f_X(x; \theta) = \frac{x}{a(\theta)} \mathbb{1}(0 \leq x \leq b(\theta)),$$

where $\theta \in \Theta = \mathbb{R}_{++}$, $a, b : \Theta \rightarrow \mathbb{R}_{++}$ are fixed functions, and $\mathbb{1}(\cdot)$ is the indicator function.

1. Consider basic distributional properties of $\mathbb{P}_\theta$.

(a) Find $a(\theta)$ and $b(\theta)$.

• **Solution:** Observe that

$$\begin{aligned} 1 &= \int_{-\infty}^{\infty} f_X(x; \theta) dx = \frac{1}{a(\theta)} \int_0^{b(\theta)} x dx = \frac{b(\theta)^2}{2a(\theta)}, \\ \sqrt{\theta} &= \int_{-\infty}^{\infty} x f_X(x; \theta) dx = \frac{1}{a(\theta)} \int_0^{b(\theta)} x^2 dx = \frac{b(\theta)^3}{3a(\theta)}. \end{aligned}$$

Solving the two equations above give $a(\theta) = \frac{9}{8}\theta$ and $b(\theta) = \frac{3}{2}\sqrt{\theta}$.

(b) Find $F_X(x; \theta) = \mathbb{P}_\theta[X_i \leq x]$, the distribution function of X_i .

• **Solution:** For $x \in [0, b(\theta)]$,

$$\int_{-\infty}^x f_X(z; \theta) dz = \frac{1}{a(\theta)} \int_0^x z dz = \frac{4x^2}{9\theta}.$$

Therefore,

$$F_X(x; \theta) = \begin{cases} 0 & \text{if } x \leq 0 \\ \frac{4x^2}{9\theta} & \text{if } 0 < x \leq \frac{3}{2}\sqrt{\theta} \\ 1 & \text{if } \frac{3}{2}\sqrt{\theta} \leq x \end{cases}$$

2. Consider the (methods of moments) estimator:

$$\hat{\theta}_m = \frac{1}{n} \sum_{i=1}^n m(X_i), \quad \text{for some chosen function } m(\cdot).$$

(a) Let $m(x) = x$. Is $\hat{\theta}_m = \bar{X}_n$ unbiased for θ ? If not, find the direction of the bias.

• **Solution:** $\bar{X}_n$ is biased for θ because $\mathbb{E}_\theta[\bar{X}_n] = \mathbb{E}_\theta[X_i] = \sqrt{\theta}$. Furthermore, it is upwards biased when $\theta < 1$ and downwards biased when $\theta > 1$.

(b) Find a choice of $m(\cdot)$ such that $\hat{\theta}_m$ is unbiased for θ . Denote this estimator by $\hat{\theta}_U$.

• **Solution:** First note that for $m(x) = x^p$ we have

$$\mathbb{E}_\theta[\hat{\theta}_m] = \int_{-\infty}^{\infty} x^p f_X(x; \theta) dx = \frac{8}{9\theta} \int_0^{\frac{3}{2}\sqrt{\theta}} x^{p+1} dx = \frac{3^p}{2^{p-1}(p+2)} \theta^{p/2}.$$

Therefore, for $m(x) = x^2$, we have $\mathbb{E}_\theta[\hat{\theta}_m] = \frac{9}{8}\theta$, which implies that

$$\hat{\theta}_U = \frac{8}{9} \frac{1}{n} \sum_{i=1}^n X_i^2$$

is an unbiased estimator of θ .

3. Consider the (maximum likelihood) estimator:

$$\hat{\theta}_{\text{ML}} = \frac{4}{9} X_{(n)}^2, \quad X_{(n)} = \max_{1 \leq i \leq n} X_i.$$

(a) Find the distribution of $\hat{\theta}_{\text{ML}}$, and show that $\hat{\theta}_{\text{ML}}$ is a complete sufficient statistics for θ .

• **Solution:** Clearly, $F_{\text{ML}}(x; \theta) = 0$ for $x < 0$. For $x \geq 0$, we have

$$\mathbb{P}_\theta[\hat{\theta}_{\text{ML}} \leq x] = \mathbb{P}_\theta \left[\frac{4}{9} X_{(n)}^2 \leq x \right] = \mathbb{P}_\theta \left[X_{(n)} \leq \frac{3}{2} \sqrt{x} \right] = \left(F_X \left(\frac{3}{2} \sqrt{x}; \theta \right) \right)^n,$$

and therefore

$$F_{\text{ML}}(x; \theta) = \begin{cases} 0 & \text{if } x \leq 0 \\ \left(\frac{x}{\theta} \right)^n & \text{if } 0 < x \leq \theta \\ 1 & \text{if } \theta < x \end{cases}.$$

The Lebesgue density of $(X_1, \dots, X_n)$ can be written as

$$\prod_{i=1}^n \frac{x_i}{a(\theta)} \mathbb{1}(0 \leq x_i \leq b(\theta)) = a(\theta)^{-n} \mathbb{1}(x_{(n)} \leq b(\theta)) \mathbb{1}(x_{(1)} > 0) \prod_{i=1}^n x_i.$$

Sufficiency directly follows from the Factorization Theorem. Moreover, for any function $g(\cdot)$ such that

$$\mathbb{E}[g(\hat{\theta}_{\text{ML}})] = \int_0^\theta g(x) \frac{nx^{n-1}}{\theta^n} dx = 0 \text{ for all } \theta.$$

After taking derivatives, it is clear that $g(\theta) = 0$ for all θ (a.s.- $\mathbb{P}$).

(b) Find a UMVU estimator of θ . Denote this estimator by $\hat{\theta}_{\text{UMVU}}$.

• **Solution:** We have

$$f_{\text{ML}}(x; \theta) = \frac{n}{\theta^n} x^{n-1} \mathbb{1}(0 \leq x \leq \theta)$$

and

$$\mathbb{E}_\theta[\hat{\theta}_{\text{ML}}] = \int_{-\infty}^{\infty} x f_{\text{ML}}(x; \theta) dx = \frac{n}{n+1} \theta.$$

Therefore,

$$\hat{\theta}_{\text{UMVU}} = \frac{n+1}{n} \hat{\theta}_{\text{ML}}$$

is a UMVU estimator of θ because it is unbiased and based on a complete sufficient statistic.

- (c) Compute $\mathbb{V}[\hat{\theta}_{\text{UMVU}}]$, and show that it is never larger than $\mathbb{V}[\hat{\theta}_{\text{U}}]$. Can these two variances be equal?

• **Solution:** Given the results above,

$$\mathbb{V}[\hat{\theta}_{\text{UMVU}}] = \frac{(n+1)^2}{n^2} \mathbb{V}[\hat{\theta}_{\text{ML}}] = \frac{\theta^2}{(n+2)n}.$$

Then, using the calculations further above,

$$\mathbb{V}[\hat{\theta}_{\text{U}}] = \frac{64}{81} \frac{1}{n} (\mathbb{E}[X_i^4] - (\mathbb{E}[X_i^2])^2) = \frac{\theta^2}{3n}.$$

Therefore, $\mathbb{V}[\hat{\theta}_{\text{UMVU}}] \geq \mathbb{V}[\hat{\theta}_{\text{U}}]$, with strict inequality for all $n > 1$. Note that if $n = 1$, then $\mathbb{V}[\hat{\theta}_{\text{UMVU}}] = \mathbb{V}[\hat{\theta}_{\text{U}}]$, but also (and crucially) $\hat{\theta}_{\text{UMVU}} = \hat{\theta}_{\text{U}}$ in this very special case.

4. Consider the following hypothesis testing problem:

$$H_0 : \theta \leq \theta_0 \quad \text{vs.} \quad H_1 : \theta > \theta_0,$$

with a test that rejects H_0 if and only if $\hat{\theta}_{\text{ML}}/\theta_0 > c$ for some critical value c .

- (a) Find c so that the test has 5% size.

• **Solution:**

$$\begin{aligned} \mathbb{P}_{\theta}[\hat{\theta}_{\text{ML}}/\theta_0 > c] &= 1 - \mathbb{P}_{\theta}[\hat{\theta}_{\text{ML}}/\theta_0 \leq c] = 1 - F_{\text{ML}}(c\theta_0; \theta) \\ &= \begin{cases} 1 - (c\theta_0/\theta)^n & \text{if } 0 \leq c < \theta/\theta_0 \\ 0 & \text{if } c \geq \theta/\theta_0 \end{cases} \\ &= \max\{1 - (c\theta_0/\theta)^n, 0\} \end{aligned}$$

for any $c > 0$ and any $\theta > 0$. Since $\mathbb{P}_{\theta}[\hat{\theta}_{\text{ML}}/\theta_0 > c]$ is non-decreasing in θ ,

$$\sup_{\theta \leq \theta_0} \mathbb{P}_{\theta}[\hat{\theta}_{\text{ML}}/\theta_0 > c] = \mathbb{P}_{\theta_0}[\hat{\theta}_{\text{ML}}/\theta_0 > c] = \max\{1 - c^n, 0\},$$

and therefore $c = \sqrt[n]{0.95}$.

- (b) Derive the power function, denoted by $\beta(\theta)$, and sketch a plot of it.

• **Solution:** We have

$$\beta(\theta) = \mathbb{P}_{\theta}[\hat{\theta}_{\text{ML}}/\theta_0 > \sqrt[n]{0.95}] = \max\{1 - (\sqrt[n]{0.95}\theta_0/\theta)^n, 0\} = \max\{1 - 0.95(\theta_0/\theta)^n, 0\}.$$

- (c) Propose a $100(1 - \alpha)\%$ (one-sided) confidence interval for θ , where $\alpha \in [0, 1]$.

- **Solution:** It directly follows from previous results that a $100(1 - \alpha)\%$ one-sided confidence interval is $[\hat{\theta}_{\text{ML}}/\sqrt[n]{1 - \alpha}, \infty)$.

Table 1.1. Discrete Distributions on $\mathcal{R}$

Uniform	p.d.f.	$1/m, x = a_1, \dots, a_m$
	m.g.f.	$\sum_{j=1}^m e^{a_j t}/m, t \in \mathcal{R}$
$DU(a_1, \dots, a_m)$	Expectation	$\sum_{j=1}^m a_j/m$
	Variance	$\sum_{j=1}^m (a_j - \bar{a})^2/m, \bar{a} = \sum_{j=1}^m a_j/m$
	Parameter	$a_i \in \mathcal{R}, m = 1, 2, \dots$
Binomial	p.d.f.	$\binom{n}{x} p^x (1-p)^{n-x}, x = 0, 1, \dots, n$
	m.g.f.	$(pe^t + 1 - p)^n, t \in \mathcal{R}$
$Bi(p, n)$	Expectation	np
	Variance	$np(1-p)$
	Parameter	$p \in [0, 1], n = 1, 2, \dots$
Poisson	p.d.f.	$\theta^x e^{-\theta}/x!, x = 0, 1, 2, \dots$
	m.g.f.	$e^{\theta(e^t - 1)}, t \in \mathcal{R}$
$P(\theta)$	Expectation	θ
	Variance	θ
	Parameter	$\theta > 0$
Geometric	p.d.f.	$(1-p)^{x-1}p, x = 1, 2, \dots$
	m.g.f.	$pe^t/[1 - (1-p)e^t], t < -\log(1-p)$
$G(p)$	Expectation	$1/p$
	Variance	$(1-p)/p^2$
	Parameter	$p \in [0, 1]$
Hyper-geometric	p.d.f.	$\binom{n}{x} \binom{m}{r-x} / \binom{N}{r}$
		$x = 0, 1, \dots, \min\{r, n\}, r - x \leq m$
$HG(r, n, m)$	m.g.f.	No explicit form
	Expectation	rn/N
	Variance	$rn m(N-r)/[N^2(N-1)]$
	Parameter	$r, n, m = 1, 2, \dots, N = n + m$
Negative binomial	p.d.f.	$\binom{x-1}{r-1} p^r (1-p)^{x-r}, x = r, r+1, \dots$
	m.g.f.	$p^r e^{rt}/[1 - (1-p)e^t]^r, t < -\log(1-p)$
$NB(p, r)$	Expectation	r/p
	Variance	$r(1-p)/p^2$
	Parameter	$p \in [0, 1], r = 1, 2, \dots$
Log-distribution	p.d.f.	$-(\log p)^{-1} x^{-1} (1-p)^x, x = 1, 2, \dots$
	m.g.f.	$\log[1 - (1-p)e^t]/\log p, t \in \mathcal{R}$
$L(p)$	Expectation	$-(1-p)/(p \log p)$
	Variance	$-(1-p)[1 + (1-p)/\log p]/(p^2 \log p)$
	Parameter	$p \in (0, 1)$

All p.d.f.'s are w.r.t. counting measure.

Table 1.2. Distributions on $\mathcal{R}$ with Lebesgue p.d.f.'s

Uniform	p.d.f.	$(b-a)^{-1}I_{(a,b)}(x)$
	m.g.f.	$(e^{bt} - e^{at})/[(b-a)t], \quad t \in \mathcal{R}$
$U(a, b)$	Expectation	$(a+b)/2$
	Variance	$(b-a)^2/12$
	Parameter	$a, b \in \mathcal{R}, \quad a < b$
Normal	p.d.f.	$\frac{1}{\sqrt{2\pi}\sigma} e^{-(x-\mu)^2/2\sigma^2}$
	m.g.f.	$e^{\mu t + \sigma^2 t^2/2}, \quad t \in \mathcal{R}$
$N(\mu, \sigma^2)$	Expectation	μ
	Variance	σ^2
	Parameter	$\mu \in \mathcal{R}, \quad \sigma > 0$
Exponential	p.d.f.	$\theta^{-1} e^{-(x-a)/\theta} I_{(a,\infty)}(x)$
	m.g.f.	$e^{at}(1-\theta t)^{-1}, \quad t < \theta^{-1}$
$E(a, \theta)$	Expectation	$\theta + a$
	Variance	θ^2
	Parameter	$\theta > 0, \quad a \in \mathcal{R}$
Chi-square	p.d.f.	$\frac{1}{\Gamma(k/2)2^{k/2}} x^{k/2-1} e^{-x/2} I_{(0,\infty)}(x)$
	m.g.f.	$(1-2t)^{-k/2}, \quad t < 1/2$
χ_k^2	Expectation	k
	Variance	$2k$
	Parameter	$k = 1, 2, \dots$
Gamma	p.d.f.	$\frac{1}{\Gamma(\alpha)\gamma^\alpha} x^{\alpha-1} e^{-x/\gamma} I_{(0,\infty)}(x)$
	m.g.f.	$(1-\gamma t)^{-\alpha}, \quad t < \gamma^{-1}$
$\Gamma(\alpha, \gamma)$	Expectation	$\alpha\gamma$
	Variance	$\alpha\gamma^2$
	Parameter	$\gamma > 0, \quad \alpha > 0$
Beta	p.d.f.	$\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1} (1-x)^{\beta-1} I_{(0,1)}(x)$
	m.g.f.	No explicit form
$B(\alpha, \beta)$	Expectation	$\alpha/(\alpha+\beta)$
	Variance	$\alpha\beta/[(\alpha+\beta+1)(\alpha+\beta)^2]$
	Parameter	$\alpha > 0, \quad \beta > 0$
Cauchy	p.d.f.	$\frac{1}{\pi\sigma} \left[1 + \left(\frac{x-\mu}{\sigma} \right)^2 \right]^{-1}$
	ch.f.	$e^{\sqrt{-1}\mu t - \sigma t }$
$C(\mu, \sigma)$	Expectation	Does not exist
	Variance	Does not exist
	Parameter	$\mu \in \mathcal{R}, \quad \sigma > 0$

Table 1.2. (continued)

t-distribution	p.d.f.	$\frac{\Gamma[(n+1)/2]}{\sqrt{n\pi}\Gamma(n/2)} \left(1 + \frac{x^2}{n}\right)^{-(n+1)/2}$
	ch.f.	No explicit form
t_n	Expectation	0, ($n > 1$)
	Variance	$n/(n-2)$, ($n > 2$)
	Parameter	$n = 1, 2, \dots$
F-distribution	p.d.f.	$\frac{n^{n/2}m^{m/2}\Gamma[(n+m)/2]x^{n/2-1}}{\Gamma(n/2)\Gamma(m/2)(m+nx)^{(n+m)/2}}I_{(0,\infty)}(x)$
	ch.f.	No explicit form
$F_{n,m}$	Expectation	$m/(m-2)$, ($m > 2$)
	Variance	$2m^2(n+m-2)/[n(m-2)^2(m-4)]$, ($m > 4$)
	Parameter	$n = 1, 2, \dots$, $m = 1, 2, \dots$
Log-normal	p.d.f.	$\frac{1}{\sqrt{2\pi}\sigma}x^{-1}e^{-(\log x - \mu)^2/2\sigma^2}I_{(0,\infty)}(x)$
	ch.f.	No explicit form
$LN(\mu, \sigma^2)$	Expectation	$e^{\mu+\sigma^2/2}$
	Variance	$e^{2\mu+\sigma^2}(e^{\sigma^2} - 1)$
	Parameter	$\mu \in \mathcal{R}$, $\sigma > 0$
Weibull	p.d.f.	$\frac{\alpha}{\theta}x^{\alpha-1}e^{-x^\alpha/\theta}I_{(0,\infty)}(x)$
	ch.f.	No explicit form
$W(\alpha, \theta)$	Expectation	$\theta^{1/\alpha}\Gamma(\alpha^{-1} + 1)$
	Variance	$\theta^{2/\alpha} \left\{ \Gamma(2\alpha^{-1} + 1) - [\Gamma(\alpha^{-1} + 1)]^2 \right\}$
	Parameter	$\theta > 0$, $\alpha > 0$
Double Exponential	p.d.f.	$\frac{1}{2\theta}e^{- x-\mu /\theta}$
	m.g.f.	$e^{\mu t}/(1 - \theta^2 t^2)$, $ t < \theta^{-1}$
$DE(\mu, \theta)$	Expectation	μ
	Variance	$2\theta^2$
	Parameter	$\mu \in \mathcal{R}$, $\theta > 0$
Pareto	p.d.f.	$\theta a^\theta x^{-(\theta+1)}I_{(a,\infty)}(x)$
	ch.f.	No explicit form
$Pa(a, \theta)$	Expectation	$\theta a/(\theta - 1)$, ($\theta > 1$)
	Variance	$\theta a^2/[(\theta - 1)^2(\theta - 2)]$, ($\theta > 2$)
	Parameter	$\theta > 0$, $a > 0$
Logistic	p.d.f.	$\sigma^{-1}e^{-(x-\mu)/\sigma}/[1 + e^{-(x-\mu)/\sigma}]^2$
	m.g.f.	$e^{\mu t}\Gamma(1 + \sigma t)\Gamma(1 - \sigma t)$, $ t < \sigma^{-1}$
$LG(\mu, \sigma)$	Expectation	μ
	Variance	$\sigma^2\pi^2/3$
	Parameter	$\mu \in \mathcal{R}$, $\sigma > 0$